ITE4116M FINAL YEAR PROJECT

# PROJECT SUGGESTION FORM

*Academic Year 2026/27*

**Submission Guide:** Students must form a group of 3 to 4 members, discuss the project topic with your preferred supervisor, and obtain verbal agreement. Once agreed, the group must complete this Project Suggestion Document and upload the document file (**in both PDF & DOCX**) on Moodle during the FYP session by **14-SEP-2026.**

## 1. General Project Information

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| **Proposed Project Title** | AI and IoT Based Food Replenishment and Waste Reduction System |
| **Assigned Project Supervisor** | Macro (proposed; verbal agreement pending) |
| **Programme / Class Code** | Higher Diploma in AI and Smart Technology (IT114124) / Class AIST-2B |
| **Student Group Number** | Group 2 |

**Student Member Details (Normally 3 to 4 members):**

| S/N | Student Name | Student ID | Email Address | Proposed Major Role / Area of Responsibility |
|---|---|---|---|---|
| 1 | Yuan Yuchen | 250342840 | 250342840@stu.vtc.edu.hk | **Project Leader / Backend AI Lead (40%)** Requirements, APIs, food recognition, demand forecasting, replenishment and integration tests. |
| 2 | Xu Ziqian | 250137555 | 250137555@stu.vtc.edu.hk | **IoT Hardware / Edge Processing (30%)** Sensor integration, Pi acquisition, calibration, filtering, tray events and hardware tests. |
| 3 | Li Wenhua | 250329909 | 250329909@stu.vtc.edu.hk | **Web Application / UX Developer (30%)** Staff dashboard, QR menu and orders, live updates, bilingual UI and usability tests. |

# 2. Problem Statement & Project Objectives

*Instructions: Describe the business or technical problem in information technology that this project intends to address. Define 2-3 specific project objectives (what your solution intends to achieve for the target users/businesses).*

**Problem Statement:**

Buffet and batch-prepared self-service outlets need timely replenishment without excess food at closing. Visual checks and separate kitchen records can leave staff unaware of remaining food and batches already in preparation, causing waiting, duplicate work and avoidable leftovers. Customer menus may also show outdated availability.

We propose an affordable prototype for continuous monitoring while trays remain in service, linking measurements, demand estimates and kitchen progress to replenishment decisions. Metafoodx and Controliza are commercial references; improvement will be evaluated against simple baselines in a small serving line.

**Project Objectives:**

1. Implement continuous monitoring for up to three serving positions, recording remaining food, refills and surface temperature while detecting unreliable readings.
2. Develop demand estimates and replenishment recommendations using preparation lead time and work in progress; compare empty-tray time and leftovers against fixed-threshold and consumption-rate baselines.
3. Integrate bilingual staff tools and a supplementary QR menu/order interface; test data consistency, interruption recovery and usability in controlled service scenarios.

# 3. Proposed Project Scope & Key Features

*Instructions: Define the scope of your system. List and briefly describe the key modules, functions, or features that your group plans to implement. Projects must be implementation-oriented and student-centered.*

**Key Subsystems / Features:**

**A. Continuous tray monitoring.** Combine weight, visual dish identification and surface-temperature readings while trays remain in service. Handle tare, movement, refills, occlusion and stale data.

**B. Replenishment and kitchen tasks.** Recommend timing and batch quantities using demand, preparation/closing times and batches in progress. Staff approve or adjust tasks; completed refills update supply.

**C. Operations dashboard and evaluation.** Show status and alerts; export empty-tray time, refill history, leftovers and disposal/reuse records for evaluation.

**D. Supplementary QR menu and ordering.** Display dish details, availability, order popularity or buffet take-up trends, and estimated waits. Support applicable fast-casual/made-to-order orders and reconcile pending orders with physical stock.

**Prototype Scope:**

Six months, one line, up to three positions and 4-6 registered dishes. Excludes payments, delivery, procurement ERP and independent food-safety certification.

**Expected Users & Roles:**
Staff monitor and refill trays; managers maintain menus/settings and review reports; customers view menus and submit or track applicable orders.

# 4. Technical Stack & Special Requirements

*Instructions: Detail the hardware and software tools you intend to use. Highlight how your project incorporates contemporary technologies (AI, IoT, Cloud, etc.) and how it aligns with modern considerations like environmental-friendly design, bilingual/multi-language support, or cross-border deployability to meet Greater China business needs.*

**Development Technologies & Tools:**
- **Hardware:** Raspberry Pi 5 with load cells and ADC interfaces (such as HX711 or NAU7802), an overhead camera, surface-temperature sensor, display and LEDs. Assess RGB-D after range and cost tests. Existing UNO R4 WiFi is optional for acquisition/debugging.
- **Programming Languages:** Python, TypeScript, HTML/CSS; C++ only if a microcontroller is used.
- **Frameworks / Libraries:** FastAPI, OpenCV, PyTorch for model training, scikit-learn, React and i18next.
- **Database / Storage:** PostgreSQL and local measurement/image logs.
- **Target Platforms:** Local Linux device and responsive staff/customer web interfaces.

**Special Project Considerations:**
- **Contemporary Technologies:** IoT sensing, local AI inference and live web updates; optional cloud backup after the local core works.
- **Environmental-Friendly Solution:** Evaluate avoidable food surplus; reuse available computing equipment and use replaceable sensing components.
- **Bilingual / Multi-Language Support:** English and Traditional Chinese interfaces.
- **Cross-Border Deployment / Greater China Needs:** Configurable language, currency and time zone, with local deployment options for later regional adaptation.
- **Feasibility:** Confirm camera/ADC compatibility, sampling stability and cost by bench tests. Use controlled service scenarios initially and arrange a field trial if access becomes available.

# 5. Endorsements & Supervisor Feedback

### Student Group Endorsement

By submitting this suggestion, we confirm that all listed group members are in agreement regarding this project concept and our assigned roles.

Student Representative Signature: _____

Date: _____

### Supervisor Vetting Decision

Supervisor, please put a ☑ in the appropriate box.

☐ **APPROVED — The group is authorized to proceed to draft their formal Project Proposal.**

☐ **REJECTED — The group must modify their project topic suggestion and resubmit for review.**

### Supervisor Feedback / Comments

### Supervisor Endorsement

Project Supervisor Signature: _____

Date: _____